

# Software Test Plan

OrangeHRM Open Source HR Management System

(Prepared per IEEE 829 Test Plan Standard)

|                               |                                                    |
|-------------------------------|----------------------------------------------------|
| <b>Application Under Test</b> | OrangeHRM Demo (opensource-demo.orangehrmlive.com) |
| <b>Document Version</b>       | 1.0                                                |
| <b>Prepared By</b>            | Vedant Balkrishna Patil                            |
| <b>Role</b>                   | QA / Software Tester                               |
| <b>Test Type</b>              | Manual + Automated (Selenium WebDriver, TestNG)    |

## 1. Test Plan Identifier

TP-ORANGEHRM-001

## 2. Introduction

This document defines the test plan for OrangeHRM, an open-source Human Resource Management System. Testing is performed against the publicly available demo instance at [opensource-demo.orangehrmlive.com](https://opensource-demo.orangehrmlive.com), which provides Employee Information Management (PIM), Leave Management, Time Tracking, Recruitment, Performance, and Administration modules. This plan is being executed as a self-directed QA practice exercise covering both manual test design and Selenium-based automation.

## 3. Test Items

| Module         | Feature(s) Under Test                              |
|----------------|----------------------------------------------------|
| Authentication | Login, Logout, field-level validation              |
| PIM            | Add Employee, Search Employee, Delete Employee     |
| Leave          | Apply Leave, Leave Type selection, date validation |
| Admin          | Add System User, User Management list, Delete User |

## 4. Features to be Tested

- Valid and invalid login authentication
- Mandatory field validation on the login form
- Adding a new employee with mandatory and optional fields
- Searching and filtering the employee list
- Deleting an employee record (modal confirmation handling)
- Applying for leave with a selected leave type and date
- Adding a new system user (Admin module) and validating role/employee binding
- Deleting a system user

## 5. Features Not to be Tested

- Recruitment, Performance, Time, and Buzz modules (out of current scope)
- Third-party/payment gateway integrations (none present in the OSS demo)
- Load, performance, and security penetration testing
- Mobile responsive/native app testing

## 6. Test Approach

Testing follows a two-layer approach:

- **Manual testing:** Test scenarios and detailed test cases are designed per ISO 29119 / IEEE 829 conventions, executed against the UI, and logged with actual results, status, and defects raised where applicable.
- **Automation:** A Selenium WebDriver + TestNG framework (Page Object Model) automates the repeatable regression-worthy flows identified from the manual test cases — login, employee CRUD basics, leave application, and user creation — using explicit waits (WebDriverWait/ExpectedConditions) and handling for alerts, modals, and dynamic AJAX loading rather than fixed sleeps.

## 7. Item Pass/Fail Criteria

A test case is marked **Pass** if the actual result matches the expected result exactly. It is marked **Fail** if the actual result deviates, an unhandled error occurs, or the application does not respond within the defined wait/timeout thresholds. A test is marked **Blocked** if a precondition (e.g., a prior dependent step) could not be completed.

## 8. Suspension and Resumption Criteria

Testing is suspended if the demo environment is unreachable, if login itself is broken (blocking all downstream modules), or if more than 50% of test cases in a module fail due to a single root cause. Testing resumes once the blocking issue is confirmed fixed or a workaround is documented.

## 9. Test Deliverables

- Test Plan (this document)
- Test Case document / spreadsheet (Test\_Cases\_OrangeHRM.xlsx)
- Selenium + TestNG automation project (source code, POM classes, config)
- Test execution summary and defect log (maintained during execution)

## 10. Test Environment

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| <b>Application URL</b>  | https://opensource-demo.orangehrmlive.com/             |
| <b>Browser(s)</b>       | Google Chrome (latest stable)                          |
| <b>Automation Stack</b> | Java 11, Selenium WebDriver 4.21, TestNG 7.10, Maven   |
| <b>Test Data</b>        | Publicly available demo credentials (Admin / admin123) |
| <b>OS</b>               | Windows / Linux (cross-platform via WebDriverManager)  |

## 11. Roles and Responsibilities

| Role                       | Responsibility                                                 |
|----------------------------|----------------------------------------------------------------|
| QA Engineer (Vedant Patil) | Test design, manual execution, automation scripting, reporting |

## 12. Schedule (Indicative)

| Activity | Duration |
|----------|----------|
|----------|----------|

|                                      |          |
|--------------------------------------|----------|
| Test planning & test case design     | 1 day    |
| Manual test execution                | 1-2 days |
| Automation framework build           | 2 days   |
| Automation execution & stabilization | 1 day    |
| Reporting & wrap-up                  | 0.5 day  |

### 13. Risks and Mitigations

| Risk                                                  | Mitigation                                                   |
|-------------------------------------------------------|--------------------------------------------------------------|
| Public demo may reset or be modified by other testers | Re-verify test data/state at the start of each run           |
| Dynamic element locators change between releases      | Centralize locators in Page Objects for single-point updates |
| Flaky waits on AJAX-heavy pages (search, save)        | Use explicit/FluentWait instead of fixed Thread.sleep()      |

### 14. Approval

Prepared by: Vedant Balkrishna Patil (QA Engineer)

Date: \_\_\_\_\_